

Capstone Project Audit: Food Safety Intelligence Platform

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Capstone Presentation Slides:

<https://docs.google.com/presentation/d/1bILs8QQu054gmXoHxqMyqvGF770bqTrCr8IZVpJGu8o/edit?usp=sharing>

Overview:

The Food Safety Intelligence Platform is a predictive analytics platform that combines restaurant inspection records, official sanitation complaints, and Yelp reviews to estimate the likelihood that a restaurant will experience a future inspection failure or critical violation. The platform serves as a solution to a high prevalence of dangerous food-borne illnesses in the United States in addition to a lack of thorough restaurant food safety risk information available to consumers. The service operates in the Chicago area and is intended for multiple users, including customers seeking safer dining options, city inspectors prioritizing inspections, and may eventually extend to businesses such as insurers or food delivery services. The platform is designed with urban public health in mind, with a niche focus on providing safety and reassurance for caregivers and immunocompromised users who would benefit from straightforward risk summaries. The platform provides predicted risk scores supported by natural language summaries and SHAP explanations, which identify the factors that contributed most to the model's prediction output.

The primary prediction target of the algorithm is the future probability that a restaurant will fail an inspection or receive a critical violation. To generate these predictions, the model combines public datasets including Chicago food inspection records, business license information, 311 sanitation complaints (public reports submitted by residents about food safety concerns), Yelp reviews, and potentially weather data. The proposed machine learning algorithm will use models such as XGBoost or LightGBM, along with natural language processing techniques to extract information from inspection and review text. Users would be able to view both the predicted risk score and the key factors contributing to the prediction through an explainability layer designed to improve transparency.

Privacy Impact Assessment:

Data Sources and Information Flows:

The datasets used by the Food Safety Intelligence Platform were originally collected for purposes other than supporting a predictive safety risk score. Inspection records, business licenses, and 311 complaint data are collected to support regulatory and public health operations. Yelp reviews are collected to help consumers evaluate businesses and share their dining experiences.

Using Nissenbaum's Contextual Integrity Framework, most of the data sources appear to be reasonably aligned with the project's public health objectives, however the Yelp reviews may present some complications. Even though the data is publicly available, reviewers may not expect their comments to be incorporated into a predictive risk scoring system. Regardless, the

general context of customers providing feedback on restaurants remains consistent with the broader context of evaluating restaurant quality and safety (Nissenbaum, 2011).

Personally identifiable information located in the data sources such as Yelp usernames or restaurant employee names would be removed or anonymized before the data is stored and processed. This aligns with data privacy principles and reduces the risk of exposing information about individuals (U.S. Department of Homeland Security, n.d.). Additionally, users of the platform would not be required to create accounts or provide personally identifiable information when using the service, however the platform may still collect location data such as ZIP codes if users make a location-based search.

Privacy Risks and Potential Harms:

Under Solove's Taxonomy of privacy, the primary privacy risks of the Food Safety Intelligence Platform relate to secondary use, aggregation, and dissemination of information (Solove, 2006). Although the source datasets are publicly available, combining inspection records, complaint data, and Yelp reviews creates new inferences that might not be readily apparent from each individual source. The platform transforms historical information into a predicted assessment of future risk, which introduces potential concerns regarding how the information could be interpreted. The capstone team plans to include a disclaimer indicating that the information presented is solely a predicted risk, and not a guarantee of future results or actual safety, which would help users make more informed decisions when given a risk score.

Although Yelp users voluntarily post their reviews publicly, they may not anticipate that their reviews will be incorporated into an algorithm used to generate restaurant risk scores. Even if the use of this public data is legally compliant, meaningful consent may not have been obtained for this specific use case. Reviewers may have personal biases and preferences reflected in their reviews that could impact a restaurant's well-being. This highlights the distinction between public availability and informed consent. Since meaningful consent may not be able to realistically be obtained from Yelp reviewers, the capstone team should review Yelp's privacy policy and terms of use to ensure that their use of review data is consistent with the expectations established by Yelp.

Under Mulligan's Privacy Analytic, several stakeholder groups may experience privacy-related harms (Mulligan et al., 2016). Restaurants can face reputational damage or lose revenue if a risk score incorrectly predicts elevated risk. Restaurant employees may also be affected if complaints or reviews contain information that identifies them. Additionally, consumers could be harmed if false negatives create a sense of safety for a restaurant that actually does pose risks that weren't captured by the predictive model. These potential harms are especially important since the platform aims to influence decision-making for dining rather than just providing information.

These potential harms could also extend beyond the restaurant itself. Reduced customer traffic as a result of inaccurate predictions could negatively impact restaurant owners and employees whose livelihoods depend on their work. These downstream effects could be disproportionately damaging for small businesses who already face barriers due to the demanding nature of the restaurant industry.

Another potential concern relates to contestability. During the interview, the capstone team indicated that restaurants would need to address incorrect information through the original data sources rather than through the platform itself. Although the team plans to provide a disclaimer explaining that risk scores are based on public data, restaurants could still face reputational or financial harms if an inaccurate prediction is displayed. Without a defined process for challenging or correcting predictions, businesses would have a limited ability to contest results they believe are unfair or inaccurate.

The platform may also create downstream harms if inaccurate predictions disproportionately affect restaurants in certain areas or customer demographics. Small businesses and restaurants located in lower-income neighborhoods could be disproportionately impacted by false positive predictions and might face reputational damage or reduced revenue.

Privacy Recommendations:

Overall, the Food Safety Intelligence Platform demonstrates awareness of several important privacy considerations, from its plan to anonymize personally identifiable information in the source data to providing transparent explanations to users. There are a few additional safeguards that could further strengthen the platform's privacy protections.

The team should establish a formal data retention and deletion protocol. The team mentioned removing personal identifiers from the source data but did not specify how long data would be stored for or how information would be removed if the source datasets were updated or corrected. This would help to improve transparency even further and reduce the risk of unnecessary data accumulation. The team should also document how data is stored and secured, especially if the platform eventually expands to include third-party integrations or partnerships.

Although the platform collects minimal information from users, they do receive location data when searches are made. While this information is not as sensitive as typical account or profile data, it could still reveal information about a user's location or dining preferences. Since location information is used, the platform should disclose how it is used and whether it is stored or retained.

To improve accountability and contestability, the team should consider implementing a formal process where restaurant owners can report inaccurate information or request review of predictions that are seemingly unfair or inconsistent with their current conditions. Even if inaccurate data needs to be corrected with the original parties that collected the data, providing a process for feedback and contesting results would increase transparency and trust for restaurants on the platform.

Model Cards:

Intended Use:

The Food Safety Intelligence Platform is intended to provide predictive risk assessments for restaurants based on publicly available historical inspection records, complaint data, and Yelp reviews. The primary intended users include consumers, city health inspectors, and business users such as insurers and food delivery platforms. The model is designed to provide data-driven support for decision-making for customers rather than a replacement for official health inspections or regulatory operations.

Per the Model Cards framework, there are out-of-scope uses of the service that should be defined for users that could pose potential harms for customers and restaurants (Mitchell et al., 2019). The risk score should not be interpreted as evidence that a restaurant is “safe” or “unsafe” to eat at, and it should not be used as the sole basis for any insurance decisions or removal from food delivery platforms. The model predicts the probability of future inspection violations but cannot guarantee actual inspection outcomes. Users should consider the risk score along with other available information to make determinations on where it is safe to dine.

Data and Model:

The capstone team plans to use established classification models such as XGBoost or LightGBM as well as a natural language processing layer to extract information from text in the source data. The model’s prediction target is the future probability that a restaurant will fail an inspection or receive a critical violation. Features used in the algorithm may include prior inspection history, complaint records, review sentiment, and potentially weather data. Since weather data is not as clearly related to restaurant food safety, the team should also document how weather contributes to the model for added transparency to restaurants and users.

The Model Cards framework emphasizes the importance of documenting data used for training algorithms because datasets can introduce preexisting biases that might be reflected in the model’s outcomes (Mitchell et al., 2019). The Food Safety Intelligence Platform relies on

historical regulatory and consumer-generated data. Inspection records and complaints could introduce biases such as underlying differences in inspection frequency or reporting behavior in certain areas or demographics. Yelp reviews could also contain subjective judgements about a restaurant that are unrelated to food safety. These additional biases could lead the model to identify patterns that don't reflect actual food safety risk.

Transparency and Interpretability:

A strength of the Food Safety Intelligence Platform is its emphasis on explainability. The platform provides SHAP explanations alongside the risk score on their webpage, showing users the features that contributed most to a restaurant's risk score. This aligns closely with transparency standards as users are better able to understand how the model came up with a prediction as well as any limitations of the predictions itself.

These explanations help improve user trust because they are able to see the underlying reasons behind the predictions, as well as allowing restaurants to better understand factors that may have contributed to a higher risk score. It is important to acknowledge, however, that these explanations alone can not guarantee accountability. Users could still misinterpret the factors presented to them. The SHAP explanations and risk scores should also come with a detailed description of how the outputs should be interpreted and what limitations they may impose.

Fairness:

One potential concern related to fairness is whether the model could systematically over-predict risk for restaurants in certain neighborhoods or with a certain customer demographic. Complaint activity and Yelp review behavior may vary across neighborhoods, and areas with more complaints could appear riskier through the model even if the actual food safety conditions are similar. This could result in the model unintentionally identifying patterns related to reporting behavior rather than actual food safety metrics. The capstone team indicated that they plan to test their model to examine this potential risk.

Yelp reviews also introduce bias because they reflect subjective customer experiences rather than actual measures of food safety. Cultural differences in table service manners and dining expectations could influence review behavior and potentially reflect poorly on restaurants that don't fit into typical American dining expectations. Yelp users are also likely a younger demographic and may not be representative of all diners, which introduces additional bias into the model.

Several groups could be disproportionately affected by inaccurate predictions, such as small businesses or restaurants located in lower-income communities. A false positive prediction could

result in lost customers by harming a restaurant's reputation and could affect the well-being of its business even without actual food safety concerns.

The team should therefore evaluate the model's performance across different neighborhoods and cuisines to ensure this potential systematic bias is not reflected in their predictions. They should also compare false positive and false negative rates across these groups to identify whether certain groups are disproportionately affected by prediction errors.

Additionally, the team should also consider how the model performs for newly opened restaurants that have limited inspection history or few customer reviews. Since there is less historical information available for these restaurants, the predictions might be less reliable. Evaluating model performance separately for newer restaurants would help to ensure they are not systematically harmed.

Accountability and Contestability:

If there are any errors in the predictions presented to users, several stakeholders could face different consequences such as a false sense of safety for immunocompromised diners, financial harms for restaurants, or misguided decisions made by future inspectors or insurers. This concern is especially important because the platform could eventually be used by food delivery platforms, and inaccurate predictions could influence business opportunities or customer access down the line even if the food safety conditions are perfectly fine.

The capstone team indicated that challenges to inaccurate information would generally need to be taken up with the original sources of data. While this approach addresses factual errors in the source data, it doesn't fully address concerns related to contestability of the model itself. Restaurants might disagree with a prediction even if all of the underlying data is technically correct and there could be contextual gaps that lead to an unfair assessment of the business.

To improve accountability, the team should implement a formal process where restaurants can submit concerns, request additional reviews of predictions, or receive more detailed explanations of the factors that contributed to a risk score. This process would help support fairness and promote stakeholder trust in the system.

Ethical Considerations:

The Belmont Principles provide a useful framework for evaluating the ethical implications of the service. Respect for persons suggests that users should be informed about how predictions are generated and what limitations are present in the model. Beneficence supports the team's goal of reducing foodborne illness and improving public health but also requires that careful attention be

paid to potential harms resulting from inaccurate predictions. Justice also requires consideration of whether there are any disproportionate burdens placed on vulnerable groups as a result of potential inaccurate predictions (Belmont Report, 1979).

The project also aligns with several Fair Information Practice Principles through the limited collection of personal data from users, anonymization of personal information in the source data, and transparency measures that provide explanations to users (U.S. Department of Homeland Security, n.d.). Additional documentation overviews data retention and procedures for correcting errors would strengthen the compliance with these principles.

Recommendations:

The capstone team should create a formal Model Card documenting intended uses, out-of-scope uses, evaluation procedures, and limitations. They should conduct fairness evaluations across demographic groups and make appropriate adjustments to their data or model to account for any unfair outcomes observed. They should also establish a formal contestability process for restaurants affected by predictions as well as providing clear explanations to users about how risk scores should be interpreted. Finally, the team should periodically re-evaluate their model performance as new data becomes available to ensure that historical biases are not amplified over time. As part of this re-evaluation effort, the team should consult relevant stakeholders such as restaurant owners, inspectors, and consumers to assess whether the platform is generating useful and fair outcomes.

Conclusion:

Overall, the Food Safety Intelligence Platform addresses an important public health problem and demonstrates thoughtful consideration of transparency through their emphasis on explainable risk scores. The use of public inspection, complaint, and Yelp review data allows the platform to identify restaurant risk of future violations, which provides useful information for consumers and health inspectors. The system also raises important privacy and fairness concerns which should be carefully considered as the model is further developed. The team should pay careful attention to the potential for biased prediction across neighborhoods and newer restaurants and the lack of a formal process for contesting inaccurate or unfair predictions. The capstone team can minimize potential harms and further improve the trust and effectiveness of the platform by more explicitly defining their data governance practices and implementing systematic fairness evaluations.

References:

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